

TECHNICAL BRIEFING · PRINCIPAL ENGINEERS & CTO

# Defense Innovation

## × Govini Ark

Integration Architecture & Engineering Value Proposition

### RDSG

Requirements-Driven  
Semantic Gateway

### SBD

Set-Based  
Design

### PSM

Predictive Scoring  
Metrics

# The Problem

Why Cold War acquisition practices fail modern defense programs

## Point-Based Design

Programs lock in a single design early, making costly rework inevitable when requirements evolve.

- Nunn-McCurdy cost breaches
- Late-stage redesign at maximum cost
- GAO: 23-year consistent failure pattern

## Static Requirements

Prose documents written once cannot represent emergent AI behaviors or volatile operational contexts.

- Attribute substitution bias
- Requirements drift vs. AI behavior
- Digital thread absent in most MDAPs

## Unquantified AI Risk

No mathematical framework exists to guarantee AI agents are calibrated or honest in mission-critical decisions.

- Hallucination with no detection
- Overconfident probability forecasts
- No auditable AI reliability metric

🔍 GAO (2025): DOD continues to struggle with delivering timely capabilities — the dominant failure mode is execution: programs proceed to commitment without completing system info-data definition.

# The Three-Pillar Solution

An integrated governance architecture that transforms Govini Ark into a high-assurance decision platform

SELF-REINFORCING GOVERNANCE ECOSYSTEM

## RDSG

Requirements-Driven  
Semantic Gateway

- SHACL structural constraints
- SWRL behavioral inference
- Requirements-Derived Query Envelopes (RDQE)
- Ontology Gap Detection
- AI Circuit Breaker pattern

## SBD

Set-Based Design +  
Agile Decision Architecture

- Deliberate design convergence delay
- MOVOL — Multiobjective Value of Information
- MBSE + RSM + Pareto pipeline
- 10–50× faster design space exploration
- Nunn-McCurdy breach prevention

## PSM

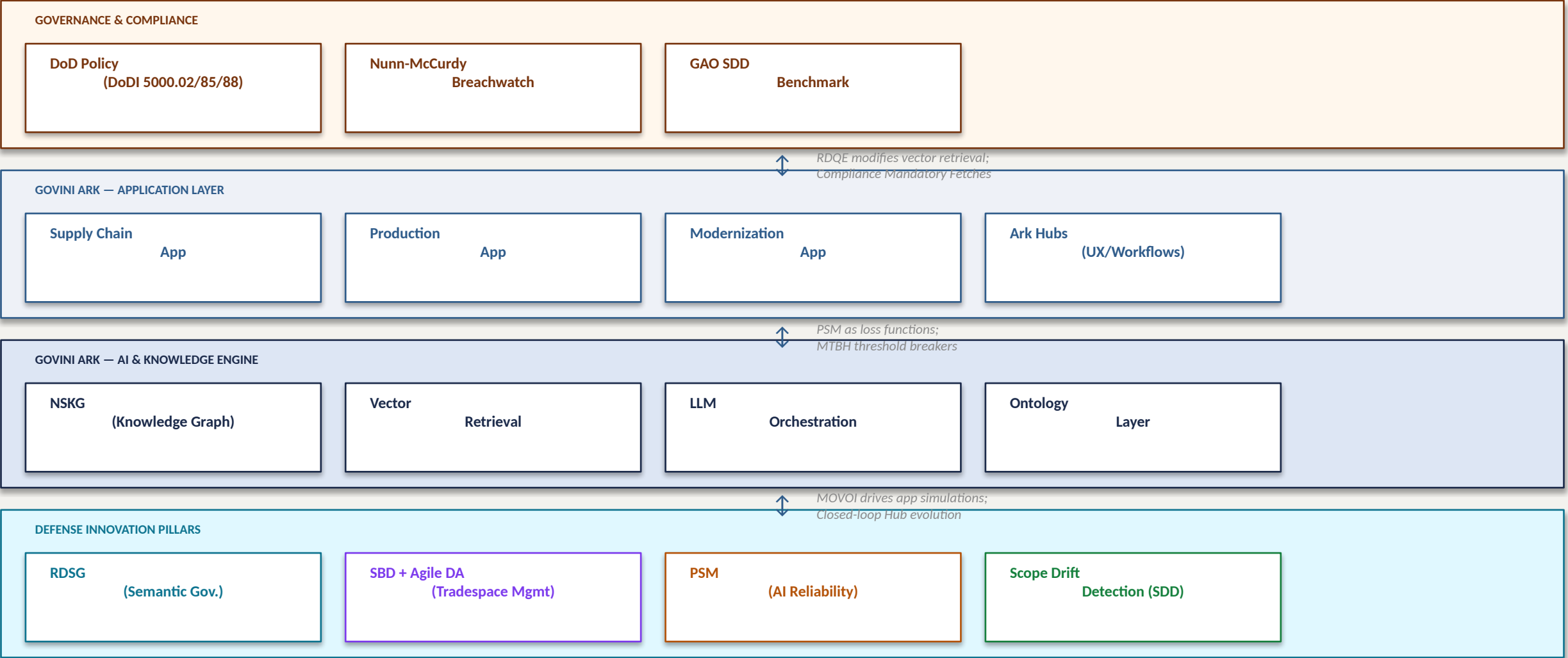
Predictive Scoring  
Metrics

- Brier Score — probabilistic calibration
- Logarithmic scoring — overconfidence penalty
- Mean Time Between Hallucinations (MTBH)
- Strictly proper scoring rule guarantees
- PM-briefable AI reliability metric

Each pillar maps directly onto Govini Ark’s existing architecture layers — RDSG → AI/Knowledge Engine · SBD/Agile DA → Application Layer · PSM → LLM training loss functions

# Integration Architecture

How the three pillars map onto Govini Ark's platform layers



# Pillar 1 — RDSG

Requirements-Driven Semantic Gateway · AI Circuit Breaker for Govini Ark

## TECHNICAL MECHANISM

### SHACL Constraints

Machine-enforceable shapes validation against ontology standards. Violations surface automatically — no human review cycle required.

### SWRL Inference Rules

Behavioral governance layer that fires inference rules when AI output violates operational requirements or crosses semantic boundaries.

### RDQE — Query Envelopes

Requirements-Derived Query Envelopes dynamically modify vector retrieval. Job Embedding Bias steers results; Compliance Mandatory Fetches force-include regulated content.

### Ontology Gap Detection

Unresolved terms are captured as OntologyGap individuals, tagged with user persona + scenario context, and queued for domain steward remediation.

## ARK INTEGRATION POINTS

- NSKG normalizes data → RDSG governs semantics
- RDQE plugs into Ark's vector retrieval pipeline
- LLM receives only ontologically compliant context
- Hub failures log → RequirementsReview auto-triggered

## ENGINEERING VALUE

### Zero-trust AI execution

Every AI directive validated before compilation

### Automated governance

SHACL/SWRL replace manual review cycles

### Live ontology evolution

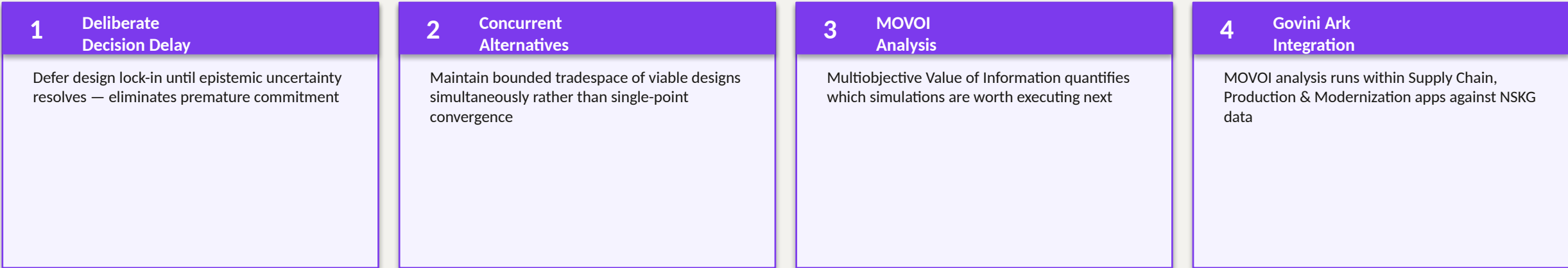
Knowledge graph updates without manual sync

### Closed-loop requirements

Operational failures become requirements signals

# Pillar 2 — SBD + Agile Decision Architecture

Set-Based Design · MOVOI · MBSE + RSM + Pareto · 10–50× Decision Velocity



AGILE DECISION ARCHITECTURE — THE DECISION PIPELINE



10–50×

faster design space exploration vs. direct simulation · Decision cycle compresses from weeks to hours · Stakeholders choose from validated Pareto frontier — not from opinion or a single design point

# Pillar 3 — Predictive Scoring Metrics

Mathematically guaranteed AI agent calibration · From qualitative assurance to auditable engineering metrics

## Brier Score

$$BS = 1/N \sum (f_i - o_i)^2$$

- Measures probabilistic calibration of AI forecasts
- A perfectly calibrated AI minimizes BS → 0
- Applied as loss function during Govini LLM training
- Penalizes both overconfident and underconfident outputs
- Provides continuous signal for model improvement cycles

## Logarithmic Score

$$LS = - \log P(\text{outcome})$$

- Strictly proper rule with infinite overconfidence penalty
- Forces agents to represent genuine epistemic uncertainty
- No adversarial strategy improves score through misreporting
- Applied to procurement decision forecasts and supply chain risk
- Decomposable into calibration + resolution components

## MTBH

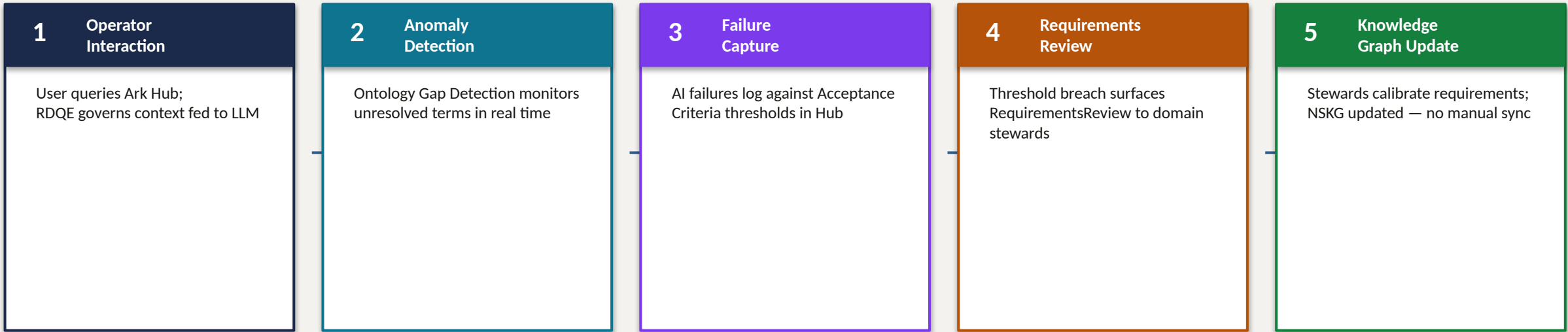
### Mean Time Between Hallucinations

- Frames AI hallucination as a reliability engineering problem
- Quantified, bounded, threshold-settable — briefable at Milestone B
- Circuit-breaker intervention when MTBH threshold violated
- Replaces qualitative 'sufficiently tested' assurances
- Standard reliability engineering methods applied to LLMs

STRICT PROPRIETY GUARANTEE · These rules mathematically guarantee AI agents report true beliefs — no adversarial strategy improves the score through misreporting.

# Closed-Loop Evolution in Ark Hubs

Transforming static dashboards into active governance tools with continuous self-reinforcement



↻ CONTINUOUS SELF-REINFORCING LOOP

## ENGINEERING IMPACT

- Domain stewards review findings directly in Hub — no separate toolchain
- Requirements evolve as operational evidence accumulates — not through manual document change
- Zero manual NSKG synchronization; governance updates propagate automatically
- AI failures become first-class engineering signals, not post-hoc audit findings

## SCOPE DRIFT DETECTION (SDD)

- Four-layer operational data capture and benchmarking
- Monitors system processes and AI behaviors in real time
- Surfaces scope drift before it propagates into info-data baseline
- Continuous quantified signal: is operational envelope drifting from formal definition?

# Standards & Regulatory Alignment

Primary realization of DoDI 5000.02, 5000.85, and 5000.88 — not compliance theater

| DoDI 5000.02                                                                                                                                                                                                                                                                    | DoDI 5000.85                                                                                                                                                             | DoDI 5000.88                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptive Acquisition Framework                                                                                                                                                                                                                                                  | Major Capability Acquisition (MCA)                                                                                                                                       | Engineering of Defense Systems                                                                                                                                  |
| <div>SBD — PRIMARY</div> <div>Deliberate decision delay preserves acquisition pathway flexibility; no full program reset required on technology insertion.</div>                                                                                                                | <div>SBD — PRIMARY</div> <div>Directly addresses the #1 MDAP failure: programs proceeding to commitment milestones without completing system info-data definition.</div> | <div>RDSG — PRIMARY</div> <div>SysML v2/KerML federated REST API delivers the 'digital authoritative source of truth' mandate at Milestone B fidelity.</div>    |
| <div>RDSG — PRIMARY</div> <div>Closed-loop requirements evolution uses AI behavioral failures as continuous refinement signals — requirements stay live.</div>                                                                                                                  | <div>PSM — PRIMARY</div> <div>Brier and Logarithmic scoring satisfy oversight requirements for AI agent reliability quantification at milestone reviews.</div>           | <div>PSM — SUPPORTING</div> <div>MTBH frames AI reliability as a standard engineering metric — quantified, bounded, and briefable at Milestone B reviews.</div> |
| <div>⚠ COVERAGE GAPS (addressable in Systems Engineering Plan): DoDI 5000.88 workforce readiness and IP strategy — structural rather than conceptual gaps; RDSG SysML v2 layer is a structural enabler, not a first-class compliance mechanism for Milestone B reviewers.</div> |                                                                                                                                                                          |                                                                                                                                                                 |

# Engineering Value Summary

What Govini Ark becomes with this integration

10–50×

Faster design  
space exploration

Zero

Manual NSKG sync  
operations

100%

Auditable AI  
reliability metrics

∞ Pen.

For AI overconfidence  
(log scoring)

| BEFORE                        | AFTER INTEGRATION                                |
|-------------------------------|--------------------------------------------------|
| ■ Manual requirements reviews | ■ Automated SHACL/SWRL enforcement               |
| ■ Point-based design lock-in  | ■ SBD tradespace with MOVOL-guided convergence   |
| ■ Qualitative AI assurance    | ■ Brier Score + MTBH — briefable at Milestone B  |
| ■ Static Ark Hub dashboards   | ■ Active governance with closed-loop evolution   |
| ■ Silent scope drift          | ■ SDD four-layer operational telemetry benchmark |
| ■ Weeks per design question   | ■ Hours — MBSE → RSM → Pareto pipeline           |

# Implementation Roadmap

Phased integration aligned with Govini Ark development cycles

| Phase 1<br>Q3 2026                                                                                                                                                                                                                                                                    | Foundation & Proof of Concept | Phase 2<br>Q4 2026                                                                                                                                                                                                                                                                   | Integration & Validation | Phase 3<br>Q1-Q2 2027                                                                                                                                                                                                                                                                                                                          | Closed-Loop Activation |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <ul style="list-style-type: none"><li>■ RDSG SHACL/SWRL schema definition against NSKG ontology</li><li>■ RDQE prototype in vector retrieval pipeline</li><li>■ PSM Brier Score baseline for existing LLM forecasts</li><li>■ SBD tradespace framework in Modernization App</li></ul> |                               | <ul style="list-style-type: none"><li>■ Full RDQE Job Embedding Bias and Compliance Mandatory Fetch</li><li>■ MTBH metric instrumented across Ark AI agents</li><li>■ MOVOI analysis live in Supply Chain and Production apps</li><li>■ Ontology Gap Detection in Ark Hubs</li></ul> |                          | <ul style="list-style-type: none"><li>■ RequirementsReview workflow surfaced to domain stewards in Hubs</li><li>■ Scope Drift Detection (SDD) four-layer telemetry operational</li><li>■ MBSE + RSM + Pareto pipeline for program decision acceleration</li><li>■ Full DoDI 5000.88 digital authoritative source of truth compliance</li></ul> |                        |
| SUCCESS CRITERIA · All three pillars instrumented · MTBH baseline established · First RequirementsReview triggered from Hub failure · Milestone B AI reliability brief delivered                                                                                                      |                               |                                                                                                                                                                                                                                                                                      |                          |                                                                                                                                                                                                                                                                                                                                                |                        |